

A top-down view of a white surface covered with a variety of mushrooms. In the top left, there's a cluster of light-colored oyster mushrooms. To their right is a dense group of shiitake mushrooms with their characteristic gills and caps. Below the oyster mushrooms is a single, dark, convex mushroom cap. In the center, a large, white, textured mushroom, possibly a king oyster, is partially visible. To the right of the center is a cluster of bright orange chanterelles. Below the center is a large, sliced mushroom showing its internal structure. In the bottom left, there's a cluster of yellow chanterelles. At the bottom, there are several more mushrooms, including some with dark caps and others with light, gilled caps.

Milkywood

Getting Started With

HOME MUSHROOM CULTIVATION

*No-waste techniques for logs,
buckets and gardens*

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No-waste techniques for logs, buckets and gardens

Fungi are a little bit weird, deeply amazing and very delicious. They are an entirely separate kingdom of life – they're neither a plant, nor an animal. And without them, life on earth might not exist.

Learning how to grow mushrooms from scratch is a bit like learning a magic trick. How do you get from a few simple supplies, to a bumper yield of gourmet or medicinal mushrooms?

And yet, once you have the basic skills and principles, you can grow your favourite mushrooms at your place.

In this guide, **we're going to get you started on your mushroom growing journey.**

We're going to introduce you to our favourite ways of growing mushrooms, take a look at what's involved, show you which techniques you can do at YOUR home and point you in the direction of additional resources to get you growing (and eating) delicious mushrooms in no time!



Pink oyster mushrooms fruiting from buckets.

Fungi are fascinating organisms – they are both complex and simple. They need certain things to grow well, and if you don't provide these, they will sulk and produce no mushrooms.

But if you simply provide them with what they need, the queendom of fungi is hugely generous and capable of producing incredible amounts of nutrient-rich mushrooms again, and again, and again – to nourish you, to nourish those around you, and ensure a functional ecosystem, too. You just need to get to know them.

So, let's get better acquainted...





You could have shiitake growing on logs down the side of your house!

Our accessible, low-waste approach to growing mushrooms

Whatever and wherever 'home' is for you, however big or small, **there is a mushroom cultivation technique that will work at your place.**

Whether you're renting an apartment or stewarding acreage, you can grow mushrooms.

The basic equipment is easily available, the processes are adaptable to your budget: it's a wonderfully accessible way to grow your own food.

Our favourite mushroom growing techniques have been adapted to make use of pre-loved and repurposed materials that we then reuse over and over again. While most commercial cultivators rely heavily on disposable, single-use plastics, **mushroom growing at home can be practically zero waste.** What a win!

Growing mushrooms is an excellent, practical way to play around with permaculture

Growing your own mushrooms is a wonderful way to turn ALL the permaculture principles into real-life lived experiences.

You can easily **Obtain a Yield** of delicious mushrooms while **Producing No Waste**. When you find yourself incubating mushrooms under your bed, or fruiting them in your bathroom, or down the shady side of your house where nothing else ever grows, you will discover that mushrooms really are the poster-child of **Using Edges and Valuing the Margins**.

And, one of our favourite things about growing mushrooms is the way they invite you to **Observe and Interact**: to tune into your home, notice the changing seasons, discover the shifting patterns of light and temperature, your climate, and the beautiful outdoors. It's an absolute delight.

By the way, you can [learn more about the 12 permaculture principles over here](https://courses.milkwood.net/mushroomcultivation), if you like.

We're Kirsten & Nick, keen mushroom growers and founders of Milkwood

We've been growing mushrooms since about 2011, and over that time we've honed our knowledge, skills, techniques and processes over thousands of grows. We have eaten a lot of home-grown mushrooms, read a lot of books and talked to many experts.

We have pioneered a bunch of low-waste, low-cost techniques that are deeply excellent for home-growing. And we shared this knowledge with 1,000+ students face-to-face, before creating our [online Home Mushroom Cultivation course](https://courses.milkwood.net/home-mushroom-cultivation).

We focus on teaching how to grow mushrooms successfully with a **low-waste, permaculture approach** - which means more mushrooms for you, and better outcomes for our planet, too.



Meet the Mushroom

Let's take a closer look at what a mushroom really is.

First, consider an apple. We know that an apple is the fruit of an apple tree. We grow apples by caring for that apple tree. When we give the tree what it needs to thrive, we get to harvest apples.

Mushrooms are very similar - **mushrooms are the fruit of a fungus**. The main body of the fungus (the "tree" bit) is called Mycelium - a strange and wonderful fluffy network of tiny filaments. The **mycelium** lives in and on a source of food called **substrate**.

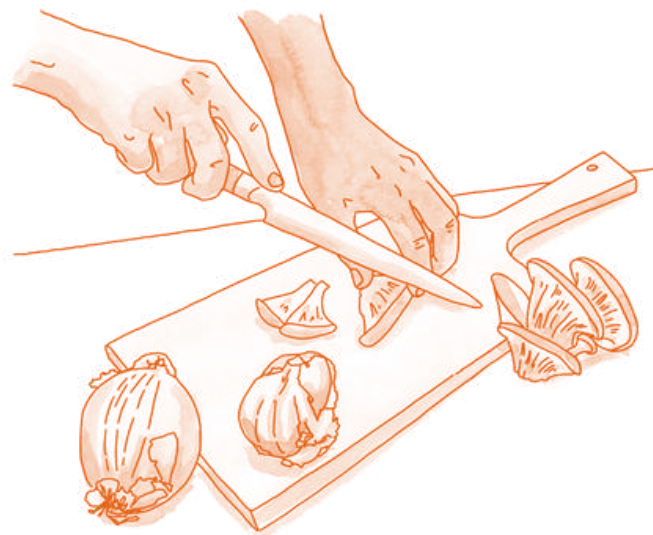


From top: Turkey tails doing their thing, beautiful yellow oyster mushrooms.

The basic mushroom-growing steps:

While each species of mushrooms like their own special blend of food (substrate), temperatures and growing environment, when you grow mushrooms at home, the steps are pretty simple:

1. **Gather supplies** - food for your mushrooms, spawn (grains or wood with mycelium already growing on them), containers or other supplies.
2. **Pasteurize and prepare your substrate** - Get rid of any nasties (like mould) that might try to steal your mushroom's food.
3. **Inoculate** - mix the grain spawn with the food.
4. **Incubate** - leave it somewhere dark, while the mycelium runs through the substrate, gathering nutrients and energy to fruit.
5. **Fruit** - in the wild, peak mushroom season is autumn, when the earth is still warm but the rains arrive and mornings are dewy and damp. To help your mycelium fruit, you change up the conditions to mimic autumn.
6. **Harvest** - in a matter of days, mushrooms arrive, ready to harvest, eat or preserve for another day.



What does this actually look like when it comes to growing at home?

Well, different mushroom species need different things in order to thrive.

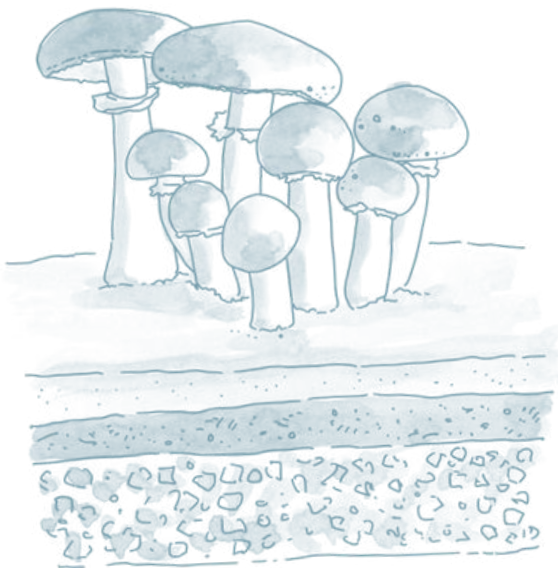
So - let's have a brief look at the overall basics of three very different approaches to growing mushrooms, to give you an idea of what's involved.



1. Growing Oyster Mushrooms in Buckets



2. Growing King Stropharia in the Garden



3. Growing Shiitake on Logs



Technique #1

GROWING OYSTER MUSHROOMS IN BUCKETS

Growing mushrooms in buckets is a great place to start, and this method is perfect for everyone - including renters and apartment-dwellers.

This technique makes it possible to have mushrooms all year round and it's a great low-waste option.

METHOD

Step 1. Gather your supplies:

You will need:

- Oyster Mushroom Grain Spawn (check our [supplier list](#)).
- Substrate (a mix of straw and sawdust)
- A collection of buckets with lids (ask your local cafe if you can have some of their food grade buckets)
- Micropore tape
- Alcohol-based sanitising spray
- A big container or clean table

Step 2. Prepare the substrate:

Mix the sawdust and straw, gather it into a cloth bag and pasteurise by placing it into a tub of hot water (60 to 80 degrees celsius) for 1 hour, before draining really well.

Congratulations - you've just pasteurised your substrate, killing off any organisms that might try to compete with your mycelium.



Inoculating substrate.

Meanwhile:

Drill holes in the side of your bucket, then cover them with micropore tape - this lets a little air in, while keeping things like mould out.

Step 3. Inoculate:

When the prepared substrate has cooled, sanitise all surfaces, and mix your grain spawn into the lovely, pasteurised straw and sawdust till it's thoroughly mixed. Pack the inoculated substrate into buckets, secure the lids and they're good to go!



Oyster mushrooms fruiting from a specially prepared bucket.

Step 4. Incubate:

Place your sealed buckets somewhere dark (under your bed or couch, or in a cupboard), where the temperature is comfortably warm for you. And wait - about 2 weeks. You can take a peek if you want to see the fluffy white mycelium growing through the substrate!

Step 5. Fruiting:

After 2-3 weeks, you'll notice the mycelium starting to bulge out of the micro-pore covered holes. Time to recreate autumn! The easiest way to mimic autumn is to move the buckets somewhere cooler, lighter and more humid: your bathroom is probably perfect!

Step 6. Harvest:

In about a week, the pins will have grown to full-sized mushrooms. To harvest, simply twist off a cluster.

Step 7. Mushrooms on toast for breakfast!

Slice, fry with butter, deglaze with a little wine, pile on a fat slice of sourdough and they're delicious. Too easy!



Technique #2

GROWING KING STROPHARIA IN THE GARDEN

If you're lucky enough to have a bit of garden, you can grow mushrooms in the great outdoors.

We love growing King Stropharia outside.

We use a two-stage technique: we start growing our mycelium in a big tub, then, after our first harvest, once our mycelium is nice and strong, we move it to a patch of earth.



King stropharia are BIG!

STAGE 1: THE TUB

Step 1. Gather your supplies:

You will need:

- Substrate: hardwood wood chips
- King Stropharia Grain spawn
- A large tub - ideally 80L (21 Gallons) - with a plug or hose at the bottom.

Timing and location:

- Choose an outdoor location that is cool, protected from wind, sun and frost (under a big tree perhaps?)
- It's best to start this grow in spring - exactly when depends on where you live, but those first frosty almost-still-winter days should be long gone, and you want to inoculate well before the heat of summer creeps up.

Step 2. Prepare your substrate:

Put your tub in its final, protected position. To pasteurise your substrate, fill the tub up with wood chips and turn on the hose.

Once full of water, leave it for a week to start fermenting, then empty out the water and you're ready to go.

Step 3. Inoculate:

Mix in your grain spawn, making sure to cover it with woodchip on top. Cover everything with some shade cloth or hessian.

Step 4. Incubate:

Within a month, you will be able to see the mycelium growing. You might be lucky enough to have fruit in as little as 6 weeks, but it's more likely the mushrooms will sprout in 6 months. During this time, make sure it gets a bit of water each week - either due to rainfall or get your hose and give it a light sprinkle.

Step 5. Fruiting:

As autumn - or spring - comes around, you can give your tub a bit of a help with fruiting. Keep it warmer on those cool nights by covering it with a plastic tarp. Or give it a casing layer of compost and coir on top. The fruit should appear within 2 weeks.



Step 6: harvesting:

While King Stropharia can grow comically big (they're also known as Garden Giants!), it's best to harvest them young and small - they taste more delicious and you are less likely to share them with pests.



From top: Making a king stropharia tub, preparing to cook a huge harvest.

Step 7: Mushrooms for lunch:

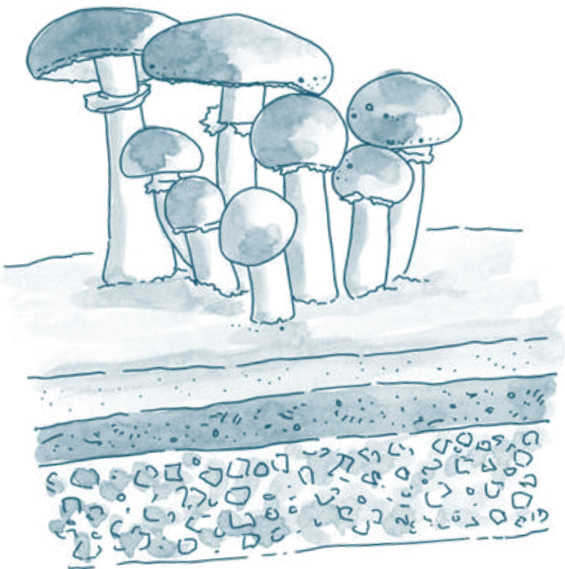
Make next-level mushroom burgers. Slice, pan fry with garlic and load onto a bun with your favourite condiments, garden greens and whatever else warms your burger-loving-heart.



From top: King stropharia fruiting in a garden, making a mushroom garden bed.

STAGE 2: IN THE GARDEN

Now your mycelium is lovely and strong, you can move it to a patch of earth, under a shady tree. Simply add another batch of pasteurised substrate into the mix under your tree, and the rest of the process will follow steps 4 to 7 (yep, the mushroom burgers are not negotiable!) and your mushroom garden V2 is underway.



Technique #3

GROWING SHIITAKE ON LOGS

Log grows are SLOW grows. With some upfront elbow grease and a little bit of patience, in 18 months to 2 years, you'll have the best shiitake you've ever eaten - and you'll get more every year for up to 5 years!



From top: Gathering logs, inoculated logs.

METHOD

Step 1. Gather your supplies:

You will need:

- Substrate: logs recently cut from healthy, living trees
- Shiitake dowel spawn (mycelium growing on little wooden dowel)
- A drill and drill bit to match your dowel
- An old pallet or two.

Timing and location:

- The best time to inoculate is early autumn, but early spring can work too.
- Find somewhere in your garden that is cool and protected from wind, sun and frost.
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Step 2. Prepare your substrate:

While we don't need to pasteurize our logs, we need to do a bit of prep. First, we give them a few weeks to rest, allowing their natural antifungal chemical defenses to dissipate. Then, we drill holes all along the logs.

Step 3. Inoculate:

Using a hammer, knock the dowel spawn into the holes till they're fully covered. Seal the inoculation points with a dab of melted wax and your mycelium is ready to go!



Shiitake logs fruiting.

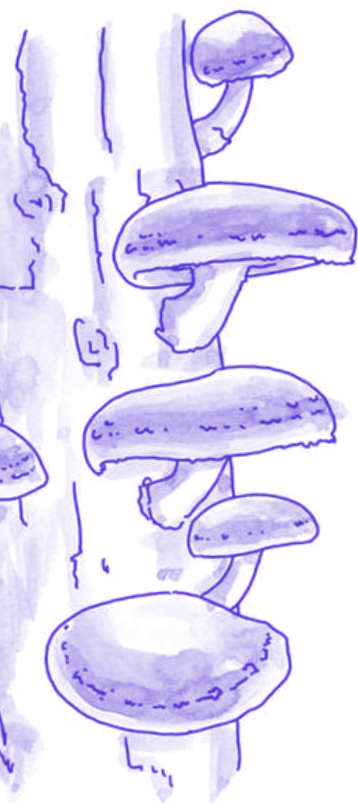
Step 4. Incubate:

Take your logs to that sheltered place in your garden, and stack them tightly on an old pallet. Cover with a shade cloth and give them a weekly drink - rain will do nicely, or on a dry week fetch the hose.

Step 5. Fruiting

Your logs are happiest fruiting in autumn, but possibly spring too, depending on where you live. After their first two summers spent eating through the log, give them a helping hand.

1. Shiitake logs appreciate a bit of a shock to jolt them into fruiting - one way is to give them a whack with a mallet, which tricks them into thinking their branch has just fallen off a dying tree and thus procreation is needed! Another is to give them a bath for about 6 hours, mimicking a big autumn rain.
2. Stack them closer to the house, in an A-frame formation so you can keep an eye on them, give them more room to fruit, and make it easy and safe for you to harvest.



Step 6. Harvesting

While you can simply twist each mushroom off, it's a good idea to slice with a sharp knife right near the bark.

When the log has finished fruiting, stack it back on its pallet at the bottom of your garden, so it can rest till it fruits next year.

Step 7: Mushrooms for dinner - with your favourite people!

Shiitake cultivation is best done with friends. It's a great idea to cultivate a shiitake 'habit' and inoculate a few more logs each year.

If you do this in autumn, consider hosting a shiitake festival - get everyone inoculating the new logs, harvest the latest batch, share big warming bowls of mushroom ramen and stir-fried shiitake and everything delicious and wonderful.

Shiitake are the perfect catalyst for building new skills, growing food, and bringing your community together.



Harvesting shiitake!



WHERE TO FROM HERE?

Growing mushrooms is fun and so rewarding. Using the bucket technique, you could be eating your own homegrown gourmet mushrooms within a month! Or, by growing shiitake on logs, you can try real, fresh, perfect shiitakes.

By now, you've probably realised that there's a fair bit to learn when getting started with your own home grows - and it can be helpful to have a place to ask all those niggling questions.

So - you might like to join our [Home Mushroom Cultivation online course](#) for expert guidance and to get all your questions answered?

Inside this course, we lead you through how to become a successful home mushroom cultivator, and get you to the point where you're growing mushrooms in buckets, in gardens, and on logs, too. We'll teach you to grow a range of both gourmet and medicinal mushrooms, to stock both your pantry and your medicine cabinet full of goodness.

Join the waitlist to be notified when our next Milkwood Home Mushroom Cultivation online course opens for bookings:

Join The Waitlist

Using simple, everyday, repurposed supplies and accessible ingredients, **it's easy to grow enough mushrooms to eat, preserve and gift to your nearest and dearest.** There's so much to love about home mushroom cultivation.



EXTRA RESOURCES

Milkwood Articles:

Check out our 15+ years of mushroom resources and articles, in our 'archive of useful things': milkwood.net/category/mushroom-cultivation

Our favourite books on Mushroom Cultivation:

- The Mushroom chapter of [Milkwood: Real Skills for Down-to-Earth Living](#), by us!
- [Growing Gourmet and Medicinal Mushrooms](#) by Paul Stamets
- [Radical Mycology](#) by Peter McCoy
- [Organic Mushroom Farming and Mycoremediation](#) by Tradd Cotter
- [Mycelium Running](#) by Paul Stamets

